THE IDEA

Ready for print starts with proof.

An evidence-backed artwork reviewer that makes source uncertainty, content discrepancies and revision regressions visible before printing.

THE PROOF CHAIN

Source fact > Requirement > Artwork evidence > Check result > Reviewer decision

The problem

Artwork checking fails when the reference is draft, the correct serving basis is unclear, or missing extracted text is mistaken for missing printed copy. A correction can also introduce a different error.

Our solution

PackSignal reads actual files, creates a source-linked truth sheet, compares supported requirements, and exposes evidence in a reviewer interface. Human decisions are recorded separately from machine results. A revised artwork triggers a full recheck.

One challenge, one focused workflow

We address “Is this Packaging Artwork ready to launch?” Source reconciliation supports this artwork-review task. We do not expand the submission into a separate brief-generation product.

Why the demo is inspectable

Official Dancing Fish files demonstrate real source ambiguity. A clearly labeled fictional Fieldday case demonstrates a quantity mismatch, missing storage copy, and a correction that accidentally removes customer-care information.

ARCHITECTURE AND MEANINGFUL AI

Interpret with AI. Verify with evidence.

The same review can run with local exact checks or optional Gemini interpretation. Provider failures remain visible; they do not create invented results. The proof viewer adds source-to-artwork trace, a zoomed evidence spotlight, PDF-calibrated cursor coordinates and a local source-hash recheck.

1. INPUTS

Artwork PDF/image + product specification + supplied rules

2. EXTRACTION

Native PDF text/coordinates, spreadsheet cells, document text

3. INTERPRETATION

Local facts + optional Gemini proposals with source quotations

4. REVIEW ENGINE

Resolve authority and basis; execute supported comparisons

5. HUMAN WORKFLOW

Inspect proof, record decisions, recheck revisions, export

Where AI earns its place

Gemini can interpret prose requirements, propose facts, read image text and raise evidence-backed questions. Exact code compares quantities, supported wording and numeric values. AI-derived quotations are checked against supplied text; interpretations and transcriptions require review.

Implementation

Node 24 HTTP server; browser HTML/CSS/JavaScript; Python PDFium, OpenPyXL, python-docx and Pillow for documents; ReportLab exports. Optional official Gemini generateContent API with structured JSON. No model-generated code is executed.

TRUST, DEMONSTRATION AND DELIVERY

Make uncertainty visible.

Four distinct outcomes

Matches reference: supported comparison agrees. Issue found: observed discrepancy or confirmed omission. Human review: authority, interpretation, claims or basis is unresolved. Unable to verify: source or extraction coverage is insufficient.

A decision is not a correction

Reviewer acknowledgment does not change the machine result. For a revised artwork, PackSignal retains the references, extracts the new file, reruns all supported checks, and identifies resolved, new, changed and not-rechecked findings.

Evidence the reviewer can inspect

Findings show source quotes, sheet/cell locations, artwork page regions, expected and observed content, and the comparison requirement. Reports include source hashes, coverage limitations, reviewer notes and revision history.

What is delivered

Working application, official organizer samples with provenance, original synthetic pack, artwork check-coverage map, red-green evidence spotlight, prepress HUD, reviewer-mapped carton context, side-by-side revision viewer, guided tour, US/India/EU starter prompts, tabular sibling-SKU screening, environmental-claim evidence prompts, exact approved-language-pair checks, source-bound designer tickets, signed HTML/PDF/JSON reports, demo script and source package.

Current validation boundary

The application has been exercised with the supplied organizer files and known fictional examples. These are not an independent accuracy benchmark. One real Gemini review of the fictional Fieldday V1 files completed using gemini-3.8-flash, returning 19 fact proposals, 12 rule proposals and 2 observations. Reviewer confirmation remains required; this single run is not an accuracy benchmark.

Scope for the judges

The map shows check coverage, not confidence. Carton faces use reviewer-drawn rectangles only for supported supplied checks; no dieline or statutory display panel is inferred. SKU and environmental signals are review prompts. Bilingual checks compare supplied exact phrases, not meaning. The prototype does not certify compliance, print production, barcode quality, translation accuracy or source authority.

Next step

Record the 3-5 minute official-sample walkthrough: proof chain, artwork map, revision regression and human-review boundary. The user performs the final recording and hackathon submission.